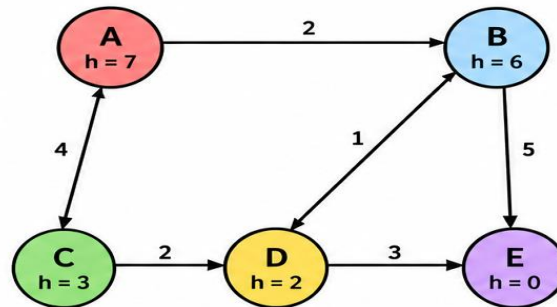


- Q1) Write a program to implement the A* Search Algorithm to find the shortest path between a source node and a destination node. [10 marks]



- Q2) Write a program to implement the K-Means Clustering Algorithm for grouping data into clusters. [20 marks]

OR

- Q2) Write a program to classify images using a Convolutional Neural Network (CNN) on the MNIST data. Normalize the input data and reshape for CNN. Split the dataset as 80:20 – train:test. Define the CNN model with 2 convolution layers, dropout of 20%, max pool size = 2X2, padding = 1, learning rate = 0.01, epochs= 10, Adam optimizer and cross entropy loss. Evaluate the model on Test Set and print its accuracy. [20 marks]

No.	Image Size	Label (Digit)	Pixel Value Range	Description
1	28 × 28	0	0–255	Handwritten digit 0
2	28 × 28	1	0–255	Handwritten digit 1
3	28 × 28	2	0–255	Handwritten digit 2
4	28 × 28	3	0–255	Handwritten digit 3
5	28 × 28	4	0–255	Handwritten digit 4
6	28 × 28	5	0–255	Handwritten digit 5
7	28 × 28	6	0–255	Handwritten digit 6
8	28 × 28	7	0–255	Handwritten digit 7
9	28 × 28	8	0–255	Handwritten digit 8
10	28 × 28	9	0–255	Handwritten digit 9

- Q3) Viva [05 marks]